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Original Research

Clinical outcome prediction using observational supervision with electronic health records and audit logs

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ABSTRACT

Objective: Audit logs in electronic health record (EHR) systems capture interactions of providers with clinical data. We determine if machine learning (ML) models trained using audit logs in conjunction with clinical data ("observational supervision") outperform ML models trained using clinical data alone in clinical outcome prediction tasks, and whether they are more robust to temporal distribution shifts in the data.

Materials and Methods: Using clinical and audit log data from Stanford Healthcare, we trained and evaluated various ML models including logistic regression, support vector machine (SVM) classifiers, neural networks, random forests, and gradient boosted machines (GBMs) on clinical EHR data, with and without audit logs for two clinical outcome prediction tasks: major adverse kidney events within 120 days of ICU admission (MAKE-120) in acute kidney injury (AKI) patients and 30-day readmission in acute stroke patients. We further tested the best performing models using patient data acquired during different time-intervals to evaluate the impact of temporal distribution shifts on model performance.

Results: Performance generally improved for all models when trained with clinical EHR data and audit log data compared with those trained with only clinical EHR data, with GBMs tending to have the overall best performance. GBMs trained with clinical EHR data and audit logs outperformed GBMs trained without audit logs in both clinical outcome prediction tasks: AUROC 0.88 (95% CI: 0.85–0.91) vs. 0.79 (95% CI: 0.77–0.81), respectively, for MAKE-120 prediction in AKI patients, and AUROC 0.74 (95% CI: 0.71–0.77) vs. 0.63 (95% CI: 0.62–0.64), respectively, for 30-day readmission prediction in acute stroke patients. The performance of GBM models trained using audit log and clinical data degraded less in later time-intervals than models trained using only clinical data.

Conclusion: Observational supervision with audit logs improved the performance of ML models trained to predict important clinical outcomes in patients with AKI and acute stroke, and improved robustness to temporal distribution shifts.

1. Introduction

Medical care can benefit from predicting patient outcomes to aid with clinical decision making, such as mortality, long length of hospital stay, hospital readmission, and sequela of renal injury [1–6]. Machine learning (ML) has shown considerable promise for clinical outcome prediction [7]. Such ML prediction models rely typically on patient data from the electronic health record (EHR) such as diagnosis codes, lab results, and vital signs. The interactions of healthcare providers with the EHR in reviewing patient data are a potentially useful, yet often neglected, source of information for training clinical prediction ML models. These interactions are captured in EHR systems as audit logs [8,9]. The audit logs record who accessed EHR data, what tasks

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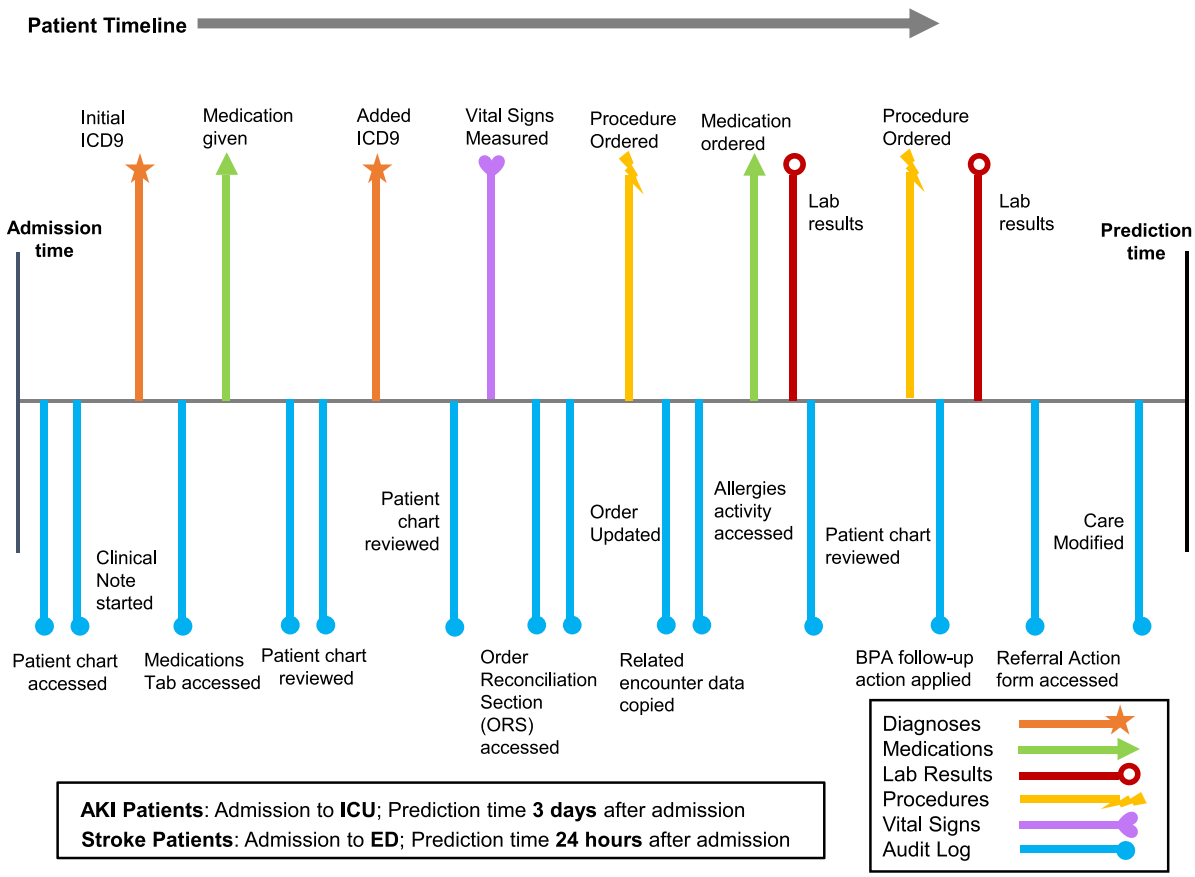

Fig. 1. An example patient timeline, with both audit log data and clinical data captured by the EHR (electronic health record) system.

they performed (typically tied to specific patient encounters), and when data access and patient events occurred (via timestamps) [8,9]. Fig. 1 shows an example of an audit log in combination with EHR data.

EHR audit logs are captured passively as healthcare personnel review and make entries in patient records. The audit logs represent "observational signals", since they arise from observing how the users interact with patient records. We refer to the paradigm of training ML models using these observational signals as "observational supervision". Incorporating observational supervision with audit log data into training clinical predication models could improve their performance.

A limitation of ML models is that they are susceptible to distribution shifts between the patient data used to train them and the patient data encountered when the models are deployed in practice [10], reducing their generalizability [10–12]. A particular challenge in clinical outcome prediction is temporal distribution shift, where the distribution of data used to train an ML model changes over time [13]. This can result in reduced model performance subsequent to its deployment. Since audit logs provide complementary information about patient conditions and care, incorporating observational supervision using audit log data into training clinical predication models could reduce the deleterious impact of temporal distribution shift on the performance of these models.

While there have been several works investigating the benefit of training ML models with audit log data in several clinical tasks [9,14,15], to our knowledge, there has been only one prior work that has explored the usage of audit log data in an outcome prediction problem [16]. This prior study showed that ML models trained using EHR audit log data can predict the likelihood of hospital discharge within 24 h, indicating that audit logs contain useful signal which can be leveraged to train an ML prediction model. However, the utility of using audit log data for training ML models to predict other clinical outcomes in specific diseases has not been explored. Moreover, the potential of observational supervision with audit log data in mitigating the reduction of performance of clinical outcome prediction models due to temporal distribution shifts in EHR data has not yet been evaluated.

In this work, we focus on clinical outcome prediction for two major patient conditions — acute kidney injury and acute ischemic stroke as exemplar diseases. Acute kidney injury (AKI) affects over 13.3 million people annually [17]. AKI occurs in more than half of the critically ill patients admitted to intensive care units (ICUs) [18,19]. The severity and duration of AKI is associated with long-term outcomes such as development of end-stage renal disease and mortality [17]. Analogous to the Major Adverse Cardiac Events (MACEs) for coronary artery diseases [20], the NIDDK (National Institute of Diabetes and Digestive and Kidney Disease) workgroup for Clinical Trials in Acute Kidney Injury put forth a composite outcome, referred to as Major Adverse Kidney Event or MAKE, to specify clinically important outcomes for AKI [21]. It includes death, new requirement of dialysis or kidney replacement therapy, and decreased estimated glomerular filtration rate (eGFR). The ability to predict a MAKE outcome in AKI patients could identify those needing closer follow-up and adjustment of renal/hypertension medications to minimize the risk of a MAKE outcome [22].

Acute ischemic stroke occurs in nearly 800,000 patients each year in the United States [23]. Some patients with stroke require hospital readmission within a short span (30 days) of discharge [24], imposing a burden on the healthcare system. The Centers for Medicare and Medicaid Services (CMS) has made reducing 30-day readmission rates a national healthcare reform goal [25]. Predicting 30-day hospital

readmission after acute ischemic stroke could help attaining this goal by identifying patients at risk and instituting appropriate management.

Previous studies applying machine learning (ML) to making clinical outcome predictions used clinical data from electronic health records (EHRs), such as diagnosis codes, medications, and lab values to determine clinical features important for prediction of MAKE-120, i.e., onset of a MAKE outcome within 120 days after ICU admission in AKI patients [19] as well as 30-day readmission after admission for acute stroke [23,26]. However, the performance of these models was limited, with low AUROCs (0.73 and 0.62 [19,23], respectively) and they were not tested for robustness to temporal distribution shifts in the data.

The objective of this study was to determine whether observational supervision with audit log data can improve the performance of ML prediction models over those trained using only clinical data in different disease outcomes. We focus on clinical outcome predictions in patients with acute ischemic stroke and AKI. We hypothesized that training ML models for clinical prediction using audit log data and clinical EHR data will improve the performance of these models compared to models trained with only clinical EHR data. We further hypothesized that observational supervision with audit log data will reduce the negative impact of temporal distribution shifts on model performance. We tested and validated these hypotheses for predicting MAKE-120 outcomes in patients admitted to the ICU for AKI and hospital readmission within 30 days after discharge in acute stroke patients.

Statement of significance

Summary	Description
Problem	Clinical outcome prediction for specific diseases using machine learning (ML) models that are trained on structured electronic health record (EHR) data may underperform when trained on limited amounts of patient cohorts and may not be robust to temporal distribution shifts in the data.
What is Already Known	Audit log data from the EHR capture interactions providers make with the EHR system as they render care and represent observational signals and can be used as complementary data to train ML models. The potential of using audit log data to improve the performance of models has been shown in a few clinical problems, such as estimating physician burnout, treatment protocol compliance, and process efficiency. However, the potential value of audit log data to improve the performance of clinical outcome prediction models for specific diseases has not been studied.
What This Paper Adds	This study investigates if training ML models with audit log data in conjunction with structured clinical data can improve the performance of clinical outcome prediction models over training the models with only clinical data in two diseases (acute kidney injury and ischemic stroke). Additionally, this study investigates whether training ML models using audit logs and clinical data improves model robustness to temporal distribution shifts that may be present in the data. The results of this study can inform further research on effectively incorporating audit log data into ML models for clinical outcome prediction in other diseases.

2. Materials and methods

2.1. Dataset and cohort

Patient data from the hospital EHR system of Stanford Healthcare between 2008 and 2018 were used for this study. We extracted structured EHR data, including demographics, diagnosis codes, lab results, procedures, medications, and vital signs. We also extracted the audit log data from the EHR. The study was approved by the Stanford Institutional Review Board. The methods and analyses to be done were designed before acquiring the data.

AKI Cohort definition. Patients who were at least 18 years old and who were admitted to the ICU for at least 24 h with an AKI diagnosis were included. ICD-9 codes of 584 (acute renal failure), 584.5 (acute renal failure with tubular necrosis), and 584.9 (acute renal failure, unspecified) were used to identify patients with AKI. Patients were excluded if they had prior kidney failure, and received renal replacement therapy or had undergone a kidney transplant before or at the time of admission, identified by their prior diagnoses codes and procedure codes. Patients who died within the first 3 days of ICU admission were also excluded. If patients had multiple ICU admissions, only their first ICU admission was included. This resulted in a cohort of 2688 AKI patients with ICU admissions between years 2014–2018 (Fig. 2A). Data prior to 2014 for these patients was not available. A summary of characteristics of the patients in the AKI cohort is provided in Appendix A, Table A.1.

Stroke Cohort definition. Patients who were at least 18 years old and who presented to the Emergency Department (ED) with an acute ischemic stroke diagnosis were included. ICD-9 codes of 433 (Occlusion and stenosis of precerebral arteries), 434 (Occlusion of cerebral arteries) and 436 (Acute but ill-defined cerebrovascular disease) were used to identify patients with acute ischemic stroke as in [27]. Patients with hemorrhagic stroke (ICD-9 codes 430-432) or transient ischemic attacks (ICD-9 code 435) were excluded. Patients who died within the hospital in the emergency room were also excluded. If patients had multiple ED admissions, only their first ED admission was included. This resulted in a cohort of 3058 stroke patients with ED admissions between years 2009–2018 (Fig. 2B). A summary of the patients in the stroke cohort is provided in Appendix A, Table A.2.

2.2. Clinical outcome prediction tasks

Outcome prediction for AKI patients. Patients who (i) died within 120 days of ICU admission, (ii) received hemodialysis or renal replacement therapy within 120 days of ICU admission, or (iii) had reduced kidney function as measured by a 50% reduction in eGFR from baseline to 120 days from ICU admission were designated as the positive class for a MAKE-120 outcome (n = 936), while the rest of the patients were designated as the negative class (n = 1809). MAKE-120 prediction was posed as a binary classification problem.

Outcome prediction for stroke patients. Patients who were readmitted to the hospital within 30 days after discharge following an acute ischemic stroke were designated as the positive class for the 30-day hospital readmission outcome (n = 1196), while the rest of the patients were designated as the negative class (n = 1862). Prediction of 30-day hospital readmission was posed as a binary classification problem.

2.3. Dataset preprocessing

Clinical data and audit log data up to 3 days after ICU admission were included for each patient in the AKI cohort for training the ML model for MAKE-120 outcome prediction as was done in [19]. Clinical data and audit log data up to 24 h after admission to the emergency department (ED) were included for each patient in the stroke cohort for training the ML model for predicting 30-day hospital readmission.

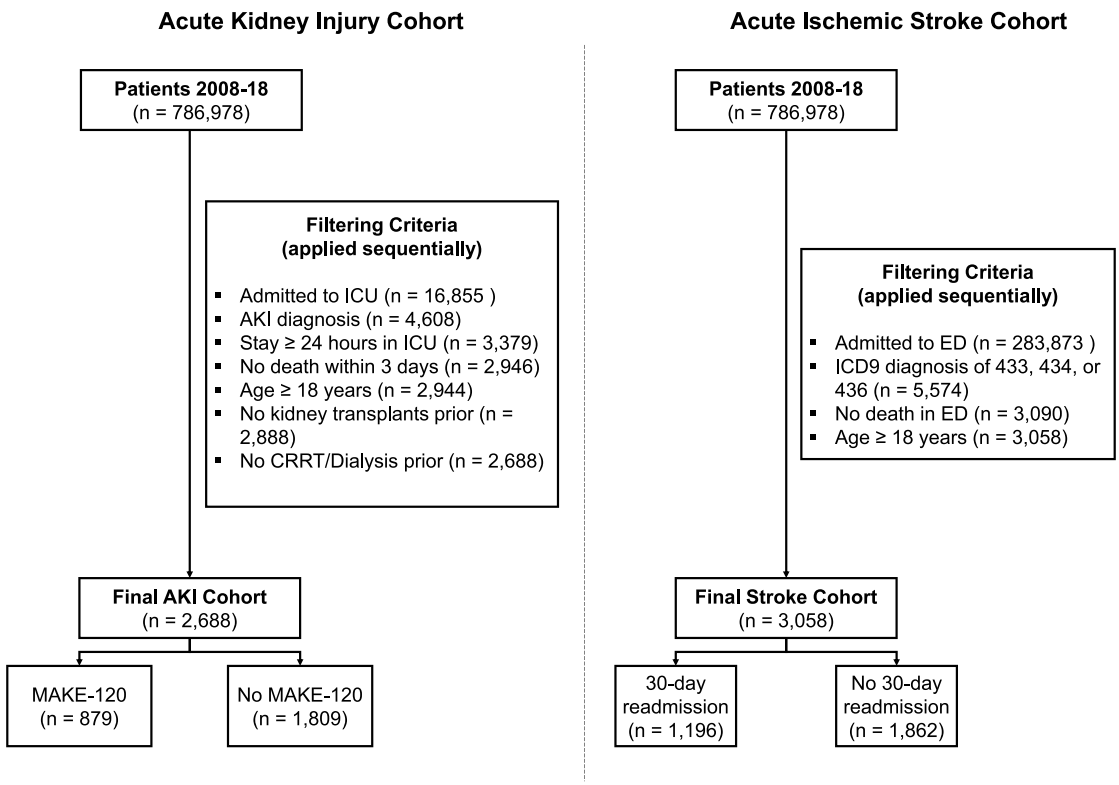

Fig. 2. Cohort Inclusion and Exclusion Criteria for (A) patients with acute kidney injury (AKI), and (B) patients with acute ischemic stroke. Abbreviations: ICU (Intensive Care Unit), ED (Emergency Department), CRRT (Continuous Renal Replacement Therapy).

Demographics (gender, race, and ethnicity), individual diagnosis codes, procedures, medication codes, and audit log events were coded as categorical values. Patient age values were binned into groups (≤ 20, 21–30 years, 31–40 years, 41–50 years, 51–60 years, 61–70 years, 71–80 years, 81–90 years, ≥ 91 years) and coded as categorical values. Net time in the ICU (for AKI patients) or ER (for stroke patients) was also included as a feature. All numerical lab values were converted to categorical variables (normal or abnormal) based on reference ranges for the lab values. EHR data that did not have a corresponding reference range were binarized as being either within the interquartile range, i.e., between the 25th and 75th percentile, or outside of it.

Each entry in the EHR and the audit log was associated with a timestamp and signified an "event". EHR events are entries made in the medical record such as orders, procedure results, and patient observations. The timestamps capture the beginning of the event. Audit log events are recorded in the EHR when EHR data are accessed, viewed, modified, or exported by providers. Some examples of clinical EHR events include 'Head CT: Order Procedure', 'Heart Rate Normal', and 'Blood Pressure Abnormal', while some examples of audit log events include 'Single Historical Immunization admin activity accessed', 'Radiology Order Confirmation Report accessed', 'E-consult message sent', and 'Patient chart reviewed'.

In this study, an encounter for a patient with AKI comprised a sequence of the events in the EHR and audit log from the time of admission to the ICU up to 3 days after admission to the ICU. Similarly, an encounter for a patient with stroke comprised a sequence of the events in the EHR and audit logs from the time of ED admission up to 24 h after admission. Events for each patient encounter were recorded as a single long sequence of "(patient identifier, event, timestamp)" tuples, ordered by timestamp. For events that occurred simultaneously, a random ordering was used. Tuples for audit log events related to billing and insurance were removed from the sequence of tuples. Nursing events were also removed. An example event sequence for a patient is shown in Fig. 1.

2.4. Featurization of the data

For each patient, we extracted features from the event sequence using bag-of-n-grams techniques [28,29], where each n-gram is a short sequence of tuples. We used these n-grams as inputs for training our machine learning (ML) models. Bag-of-n-grams extracts small segments of sequences from the long sequence of tuples that represent a patient encounter. We used n-gram sequences of varying lengths (n = 1 (unigram), 2 (bigram), 3 (trigram)), that were concatenated for generating features of the clinical EHR data and audit log data. Fig. 3 illustrates bag of n-grams featurization for events in the event sequence from the example patient timeline given in Fig. 1. To reduce the dimensionality of the data, features that occurred in < 1% of the patients in the cohort were removed for the AKI and stroke cohorts. This resulted in a total of 11,564 and 10,047 features that were used for training the AKI and stroke outcome prediction models, respectively. Of these features, 5099 and 4812 features were clinical features for the two cohorts respectively, while the rest of the features contained audit log events. Out of the 11,564 features used for training the AKI outcome prediction models, 3089 features were unigrams, 6102 were bigrams, and 2373 were trigrams. Out of the 10,047 features used for training the stroke outcome prediction models, 2995 features were unigrams, 4002 were bigrams, and 3050 were trigrams.

2.5. Model training and evaluation

We trained several different ML models for predicting the outcomes in AKI and stroke patients: logistic regression, random forests (RFs), support vector machine (SVM) classifiers, neural networks (multi-layer perceptrons (MLPs)), and gradient boosted machines (GBMs). We used the scikit-learn [30] library to train all the classifiers and implement logistic regression, RFs, SVMs, and MLPs. We used the the LightGBM [31] library for implementing GBMs and used the logistic loss for outcome prediction in our patient cohorts. We trained these models with and

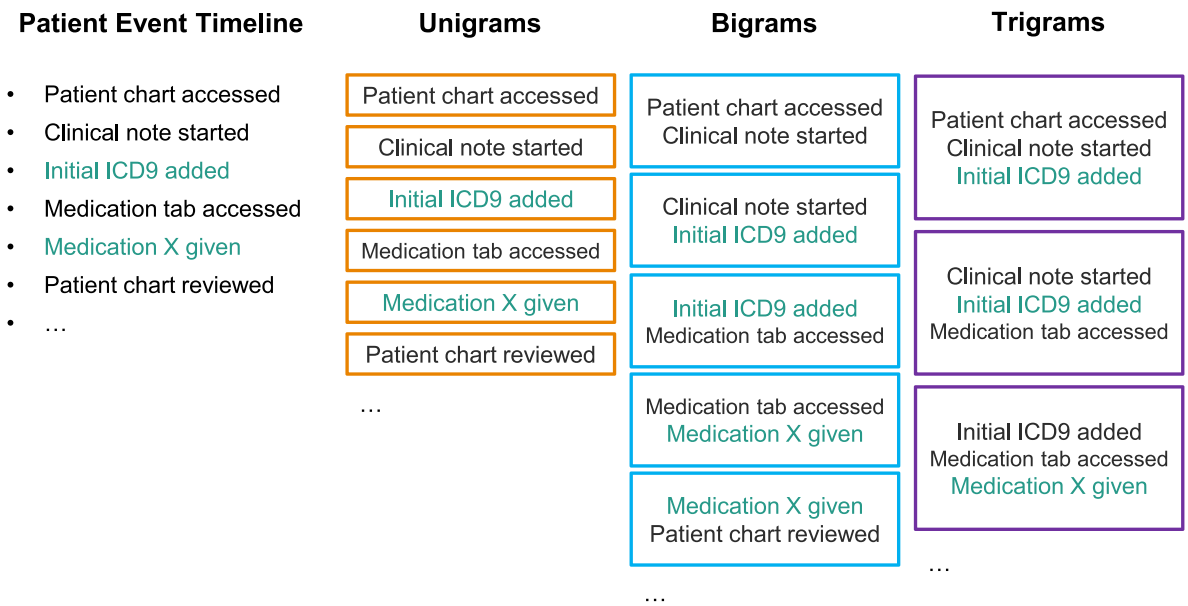

Fig. 3. Bag of n-grams for the example patient event timeline in Fig. 1. Only some of the events are shown. Events marked in green denote clinical EHR events while other events denote audit log events.

without observational supervision, i.e., with and without audit log data. We used an 80%–5%–15% training–validation–test split for patients in both the AKI cohort and the stroke cohort. Hyperparameter tuning and model optimization were done using five-fold cross-validation with the training and validation splits. The test split served as held-out data to test the final model. The models having the best average AUROC after training and hyperparameter optimization for the AKI and stroke cohorts were selected as the final models. The hyperparameters for the final models with the best average AUROC for AKI and stroke outcome prediction are listed in Appendix B.

We evaluated all of the models on the held-out test set using the area under the receiver operating characteristic (AUROC) curve. The receiving operating characteristic (ROC) curve indicates the discriminative ability of the binary classification models and was generated by varying the classification thresholds. Mean and standard deviations of the AUROC were estimated using 1001 bootstrap samples of the held-out test split, as was done previously [32]. We also plotted the calibration curve for the models with best average AUROC for the AKI and stroke cohorts.

2.6. Measuring robustness to temporal distribution shifts

To evaluate the effect of potential temporal data distribution shift on ML model performance for AKI patients, we split the AKI patient data between 2014–2018 into three time intervals: years 2014–15, 2016–17, and 2018, resulting in 1016, 1238, and 461 patients in the three date ranges, respectively (data for these patients prior to 2014 were not available). We trained the model having the best average AUROC from Section 2.5 on data from patient encounters in the 2014–2015 time interval and tested the performance on data from encounters during the two subsequent time intervals. To evaluate how well the model performs in the setting of potential temporal distribution shift in the data for the stroke patients, we split the stroke patient data between 2009–2018 into three time intervals: years 2009–12,2013–2014, and 2015–2018, resulting in 1558, 904, and 596 patients in the three date ranges, respectively. We trained the model having the best average AUROC from Section 2.5 on data from patient encounters in the 2009–2012 time interval and tested the performance on data from patient encounters during the two subsequent time intervals. For both patient cohorts, we trained the models with an 80%-5%–15% training–validation–test splits with five-fold cross-validation on the first time interval (2014–15 for the AKI cohort and 2009–12 for the stroke cohort) and evaluated the model on data from patient encounters in the subsequent date ranges using the mean and standard deviation of the AUROC, estimated using 1001 bootstrap samples. Statistical significance of differences in AUROC was evaluated using the non-parametric DeLong's test [33]. A threshold of 0.05 was used to determine significance.

2.7. Identifying informative features

In addition to assessing prediction performance, we evaluated the feature importance of all features in the models having the best average AUROC. We calculated the SHapley Additive exPlanations (SHAP) value [34] for each feature in the model for models trained with clinical EHR data and audit log data for AKI and stroke patient cohorts. We then identified the most important features contributing to each model's predictions by ranking the features based on their SHAP values.

3. Results

3.1. Overall performance

Outcome prediction for AKI patients. The AUROC for all ML models trained with and without audit logs for MAKE-120 prediction are reported in Appendix D. Overall, the performance of models trained with audit logs was better in terms of average AUROC than their counterparts trained without audit logs. GBM models tended to have better AUROC than the other ML models and were chosen as our best performing models for MAKE-120 prediction. The GBM model trained with audit log data and clinical data performed significantly better (AUROC 0.88, 95% CI: 0.85, 0.91) in MAKE-120 prediction on held-out test data compared to the same model that was trained using only clinical data (AUROC 0.79, 95% CI: 0.77, 0.81). Fig. 4 shows the ROC and calibration plots for the GBM models trained with and without audit log data. The model trained using audit log data was better calibrated than the model trained without audit log data for MAKE-120 prediction.

Outcome prediction for stroke patients. The AUROC for all ML models trained with and without audit logs for 30-day hospital readmission prediction are reported in Appendix D. Overall, the performance of models trained with audit logs was again better in terms of average

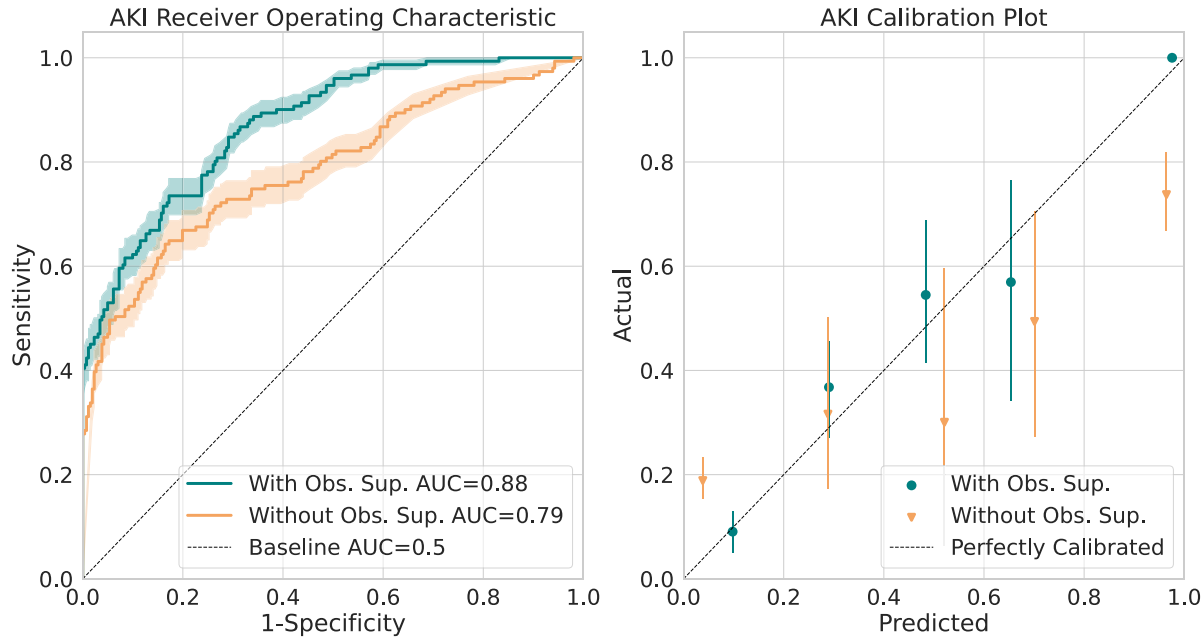

Fig. 4. (left) ROC curves, and (right) Calibration curves, for MAKE-120 outcome prediction with and without observational supervision using audit log data (Obs. Sup.) in AKI patients.

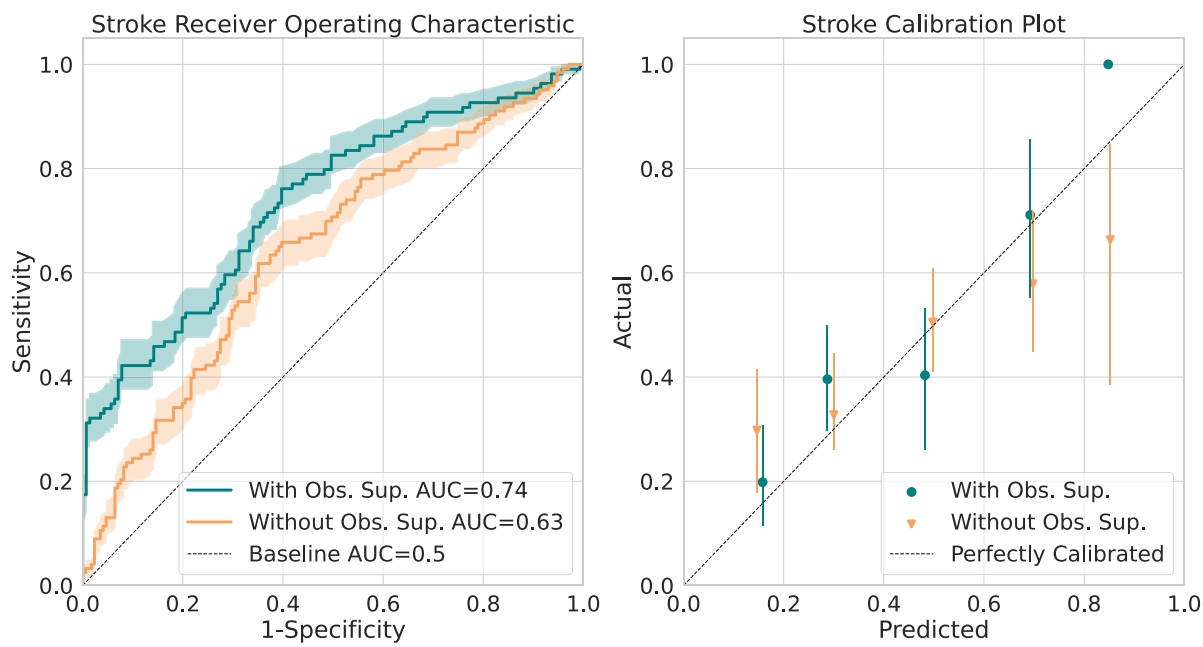

Fig. 5. (left) ROC curves, and (right) Calibration curves, for 30-day hospital readmission prediction with and without observational supervision using audit log data (Obs. Sup.) in acute ischemic stroke patients.

AUROC than their counterparts trained without audit logs. GBM models tended to have better AUROC than all of the other ML models and were chosen as our best performing models for 30-day hospital readmission prediction. The GBM model trained with audit log data and clinical data performed significantly better (AUROC 0.74, 95% CI: 0.71, 0.77) in predicting 30-day hospital readmission on held-out test data compared to the same model that was trained using only clinical data (AUROC 0.63, 95% CI: 0.62, 0.64). Fig. 5 shows the ROC and calibration plots for the GBM models trained with and without audit log data. Models trained with and without audit log data were comparable in calibration for 30-day hospital readmission prediction.

3.2. Benefit of audit log data for improving model performance

The models trained using both clinical EHR data and audit log data had more features than those that were trained only using clinical EHR data. To exclude the possibility that the improved results seen by training the models with audit log data and clinical data are simply due to increase in number of features, we also trained models for the AKI and stroke prediction tasks using clinical EHR data in which no events were removed. The number of features used in this model was greater than the number of features used in clinical only model for the AKI and stroke use cases (Table 1). We found that the prediction models

Table 1
Vocabulary sizes and AUROC for (a) model trained with clinical EHR data only with all events included, and (b) model trained with clinical EHR data only with low frequency events (<1%) removed.

	Clinical EHR model (all events included)		Clinical EHR model (<1% frequency events removed)	
	Number of features	AUROC (95% CI)	Number of features	AUROC (95% CI)
MAKE-120 prediction in AKI patients	9110	0.71 (0.68, 0.73)	5099	0.79 (0.77, 0.81)
30-day hospital readmission prediction in stroke patients	8171	0.57 (0.54, 0.60)	4812	0.63 (0.62, 0.64)

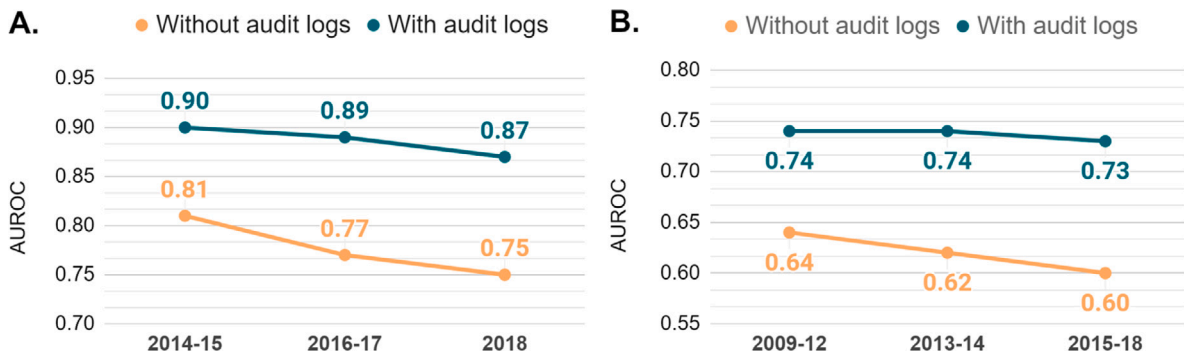

Fig. 6. Training with audit log data (observational supervision) makes ML models more stable over time for (A) MAKE-120 prediction in AKI patients, and (B) 30-day hospital readmission prediction in acute stroke patients.

trained using all of the clinical events performed worse for predicting MAKE-120 and 30-day hospital readmission than those trained with fewer clinical events for both the AKI and stroke cohorts, respectively (Table 1), suggesting that an increased number of features did not result in the models being more robust.

3.3. Robustness to temporal distribution shifts

Similar to [13], we found that the performance of the GBM models trained on only EHR data and the performance of models trained on both audit log data and EHR data degrade with time due to temporal distribution shift in the data (Fig. 6, Table 2). However, the performance degradation is lower in models trained with both EHR data and audit log data for both of the patient cohorts. We found that the model predicting MAKE-120 for AKI patients trained with both EHR and audit log data performed better when tested on data acquired in later time intervals than the model trained with only EHR data (Fig. 6A and Table 2, top). The AUROC evaluated on later time intervals show no statistically significant degradation when trained with audit log data. Similarly, we found that the model predicting 30-day readmission for stroke patients trained with both EHR data and audit log data performed better on subsets of data acquired in later time intervals than the model trained with only EHR data (Fig. 6B and Table 2, bottom). Again, AUROC evaluated on later time intervals show no statistically significant degradation when trained with audit log data. These results suggest that including audit log data in the training of the prediction models made them more robust to temporal distribution shifts with overall better performance.

3.4. Identifying informative features

We analyzed the ML models to identify the most informative features for clinical outcome prediction using their SHAP values (Section 2.7). Appendix C lists the top 50 features for MAKE-120 prediction and 30-day hospital readmission prediction for models trained using both clinical EHR data and audit log data. Several of the clinical features are clinically relevant to the predicted outcome. For example, Calcium Chloride, typically used to treat hyperkalemia or deficiency of potassium, is associated with AKI outcomes [35], while RDW (RBC distribution width), BUN (blood urea nitrogen), and length of stay in the ICU are associated with patient mortality in AKI patients [36–38]. Similarly, age and heart rate are associated with stroke outcomes [39], while hypoxia (low SpO2) and vomiting (typically treated with Ondansetron medication) are predictors of stroke outcome [40, 41]. Moreover, we found that 21 of the top 50 features for MAKE-120 prediction in the AKI patients and 40 out of the top 50 features for 30-day hospital readmission prediction in the stroke patients (Appendix C) were audit log features. The relevance of these audit log features to the clinical outcomes is not clear since they are not granular (e.g., Report reviewed, Imaging accessed). However, given that a substantial number of audit log events were informative, this suggests that they provide complementary information to the clinical features and that they contain useful signal for outcome prediction.

4. Discussion

In this work, we investigated the benefit of training clinical prediction models with observational supervision using audit log data and clinical data in the EHR. Prior works demonstrated the value of leveraging audit logs in accomplishing several tasks, such as understanding hospital workflows [15], understanding treatment care processes [42], identifying physician burnout [14], and studying clinical activity [9].

We studied the benefit of training ML models with audit log data in combination with clinical EHR data to predict two different clinical outcomes, MAKE-120 outcomes in AKI patients and 30-day hospital readmission after discharge in patients with acute ischemic stroke. These particular outcomes and diseases were selected because of their importance and impact on the patient population. This works shows the importance of the incremental benefit of using audit logs, compared with using only clinical EHR data for training ML models for predicting outcomes in specific diseases. The audit logs record interactions providers have with the EHR, and thus comprise of observational signals that can enhance the performance of clinical prediction models in this paradigm we call observational supervision. We found that models

Table 2
Robustness to temporal distribution shifts in (top) MAKE-120 prediction in AKI patients, (bottom) 30-day hospital readmission prediction in acute ischemic stroke patients. **Bold** indicates statistical significance, i.e., p-value < 0.05.

AKI patients: Evaluation Data AUROC (95% CI)						
Input Data	Training Data Range	AUROC (2014–15)	AUROC Difference (2016–17)	p-value	AUROC Difference (2018)	p-value
Clinical EHR	2014–15	0.81 (0.79, 0.83)	**0.04 (0.012, 0.068)**	**0.006**	**0.06 (0.032, 0.088)**	**<0.001**
Clinical EHR + Audit log	2014–15	0.90 (0.88, 0.92)	0.01 (−0.012, 0.032)	0.387	0.03 (−0.006, 0.066)	0.103
Stroke patients: Evaluation Data AUROC (95% CI)						
Input Data	Training Data Range	AUROC (2009–12)	AUROC Difference (2013–14)	p-value	AUROC Difference (2015–18)	p-value
Clinical EHR	2009–12	0.64 (0.62, 0.64)	0.02 (−0.002, 0.402)	0.079	**0.04 (0.018, 0.062)**	<0.001
Clinical EHR + Audit log	2009–12	0.74 (0.73, 0.75)	0.00 (−0.014, 0.014)	1.000	0.01 (−0.012, 0.032)	0.388

trained with both audit log data and clinical EHR data performed better in predicting outcomes than models trained with only clinical EHR data in both of our clinical use cases — AUROC 0.88 (95% CI: 0.85, 0.91) vs 0.79 (95% CI: 0.77, 0.81) for predicting MAKE-120 in AKI patients, and AUROC 0.74 (95% CI: 0.71, 0.77) vs 0.63 (95% CI: 0.62, 0.64) for predicting 30-day readmission in patients with acute ischemic stroke. As a historical comparison, a prior study (with current best performance) [19] using clinical data alone reported 0.73 AUROC for MAKE-120 prediction. Prior studies from literature using clinical data alone, with current best performance, reported 0.62 (95% CI: 0.61, 0.63) AUROC [23], 0.74 (95% CI: 0.68, 0.78) AUROC [26], and 0.70 (95% CI: 0.62, 0.78) AUROC [27] for 30-day hospital readmission prediction. Although the feature sets, clinical variables, and patient data used in each of these studies was different from that in the current study, our results suggest there is potential value in observational supervision with audit log data in training clinical prediction models.

Another novel aspect of our work was studying whether training ML prediction models with audit log data and clinical EHR data can improve robustness of these models to the presence of temporal data distribution shifts. We found that temporal data distribution shift occurred in the two clinical scenarios we studied, reducing the model performance for predicting outcomes (Fig. 6). Temporal data distribution shifts are well known to occur in medical data that reduce the performance of ML models, and a variety of methods have been studied to address this problem [43]. We found that the performance of models trained using audit log data reduced the degradation of performance of these models when evaluated on patients in subsequent time intervals (Fig. 6). Thus, observational supervision using audit log data produced models that were more robust to temporal distribution shifts. We believe these results are due to the fact that audit logs capture user-EHR interactions during clinical care that are complementary information to clinical data in the EHR, hence improving model robustness. While this work demonstrated data distribution shift in the two specific clinical outcome prediction problems we studied, patient data for all diseases and outcomes would likely be affected by temporal data distribution shifts that could be mitigated by our approach.

We analyzed the models to identify the most informative features used to make the predictions. We found that many of the top 50 informative features are audit log features (Section 3.4). This suggests that the audit logs capture relevant information about the patient's condition that improves the performance of the ML models. The computed SHAP scores (Appendix C) provide insights about the features that contribute most to the model prediction and provide explainability to practitioners in evaluating model predictions. Interestingly, the top 50 informative features were all unigram features. This could be the result of the large feature space of bigrams and trigrams (over 7500 unique features on average) but relatively low frequency of occurrence of these features across patients.

We found that Gradient Boosted Machines (GBMs), implemented using the LightGBM [31] framework, tended to perform the best when compared to other ML models including logistic regression, support vector machines, multi-layer perceptrons, and random forests (Appendix D). This has been observed in literature as well where GBMs have been shown to outperform both classical ML techniques like logistic regression [44] and deep learning networks when trained with structured data [45]. There may be value in using higher capacity sequence-to-sequence model architectures, such as transformers, and other training strategies, such as self-supervision, to further improve performance in future work. Regardless, we believe that incorporating audit log data in training ML prediction models will be useful for improving their performance.

There are clinical and research implications to this study. Making models more robust to distribution shifts is important for ensuring stability in model performance over time and to improve clinical utility. The findings also suggest potential generalizability to other diseases and outcomes. Research implications include showing the informative audit log features to the clinicians that could provide explainability of the model. Another research direction is incorporating additional observational signals such as eye gaze, in addition to audit logs, that may improve performance.

The audit logs capture the interactions of providers with clinical data, and thus, would not provide more PHI about the patient than the patient data themselves. However, the recorded interactions of providers with the data could also be considered to be data that should be subject to similar patient privacy and data security provisions as PHI data. Accordingly, ML models that use audit log data will need to be deployed in systems that adhere to similar data privacy and security measures as used by EHR systems. Preferably, such ML models would be ultimately simply be incorporated into EHR systems.

There are some limitations in the study. We used only structured EHR data in order to reduce the complexity of the data representation. In future work, it could be useful to generate feature representations from narrative clinical reports and medical images to explore whether these types of data improve the performance of the prediction models. While we have evaluated our approach in two different diseases and focused on two specific clinical outcomes that were chosen for their clinical importance, a second limitation is that we did not evaluate our methods for predicting outcomes in other diseases. Given the differences in diseases and outcomes, we believe our results will generalize, and the utility of our approach for training prediction models across a broader range of clinical outcomes for additional diseases could be undertaken in future work. Acquiring larger amounts of clinical data would also be helpful to confirm our results that show the benefit of using audit log data to address the challenge of temporal data distribution shift. A third limitation is that the patient data in this study were collected from a single site. Evaluation of our approach with EHR and audit log data from other centers that have different kinds of audit log data would be helpful to confirm our results and the generalizability of our findings.

5. Conclusion

Observational supervision using audit log data from our EHR system improved the performance of ML prediction models compared to models trained on clinical data alone in two different clinical outcome prediction tasks, MAKE-120 outcome prediction in critically ill patients with AKI and 30-day prediction of hospital readmission in patients with acute ischemic stroke. Observational supervision with audit log data also made the prediction models more robust to temporal distribution shift in the data. Audit log data may be useful to improve the performance of predictive ML models in other clinical scenarios.

CRediT authorship contribution statement

Nandita Bhaskhar: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Wui Ip:** Conceptualization, Supervision, Writing – review & editing. **Jonathan H. Chen:** Supervision, Data curation, Writing – review & editing. **Daniel L. Rubin:** Conceptualization, Supervision, Funding acquisition, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jonathan H. Chen reports financial support was provided by Gordon and Betty Moore Foundation. Wui Ip reports a relationship with nference, Inc. that includes: employment. Co-author Jonathan H Chen (JHC) receives research grant support from NIH/National Institute on Drug Abuse Clinical Trials Network (UG1DA015815 - CTN-0136), Stanford Artificial Intelligence in Medicine and Imaging - Human-Centered Artificial Intelligence (AIMI-HAI) Partnership Grant, Doris Duke Charitable Foundation - Covid-19 Fund to Retain Clinical Scientists (20211260), Google, Inc. Research collaboration Co-I to leverage EHR data to predict a range of clinical outcomes, American Heart Association - Strategically Focused Research Network - Diversity in Clinical Trials. JHC reports being Co-founder of Reaction Explorer LLC that develops and licenses organic chemistry education software. Was paid consulting fees from Sutton Pierce and Younker Hyde MacFarlane PLLC as a medical expert witness.

Data and code availability

The electronic health records data and audit log data used this article was provided by the Stanford Medicine Research Data Repository. All code used in this work will be made publicly available at https://github.com/nanbhas/ObservationalSupervision.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jbi.2023.104522.

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